SYNOPSIS

Database Management System - Online Shopping Management System

1. Project Title

Online Shopping Management System

2. Introduction & Objective

The Online Shopping Management System is a command-line interface (CLI) application designed to automate e-commerce operations. Built using Python 3.x and SQLite3, the project aims to demonstrate fundamental database management principles, including relational table design, CRUD operations, role-based access control, secure authentication, and robust exception handling.

3. Scope & Key Features

- **User Authentication & Authorization**: Supports secure user registration and login functionality. Implements Role-Based Access Control (RBAC) supporting **Admin** and **Customer** roles with restricted permissions. The database comes pre-loaded with **10 users** for immediate testing.
- **Product Catalog Management (CRUD)**: Administrators can perform full CRUD operations—adding new products, viewing the catalog, updating prices and stock levels, and deleting items.
- **Order Processing & Transactions**: Customers can browse available products, check live stock quantities, place orders securely, and view their individual purchase history.
- **Data Integrity & Consistency**: Foreign key constraints maintain relational integrity between users, products, and orders. Transactions ensure automatic stock deductions upon successful order placement.

4. Database Architecture & Schema Design

The SQLite database (online_shopping.db) consists of three core related tables:

- **users Table**: Stores user credentials, roles (Admin or Customer), and contact info (user_id as Primary Key).
- **products Table**: Maintains inventory data including item names, descriptions, prices, and stock counts (product_id as Primary Key).
- **orders Table**: Tracks purchase logs, linking customers to products via Foreign Keys (order_id as Primary Key, user_id and product_id as Foreign Keys).

5. Technical Stack

- **Programming Language**: Python 3.x
- **Database**: SQLite3 (using the built-in Python sqlite3 module)

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